

GHIBRAN ESA ANUGRAH

Telecommunications Engineering | Politeknik Negeri Semarang |
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PROFESSIONAL SUMMARY

Telecommunications Engineering graduate from Politeknik Negeri Semarang with hands-on experience spanning fiber optic infrastructure, RF engineering, embedded systems, and wireless communications. Certified RF Engineer and Cisco CCNA trained, with a proven track record in FTTx troubleshooting, 144-core feeder cable restoration, ESP32-based IoT development, and antenna simulation. Proficient in combining hardware-level system integration with network-level technical operations. Seeking a role as a Telecommunication Engineer, Field Engineer, Embedded Systems Developer, or RF Engineer.

EDUCATION

POLITEKNIK NEGERI SEMARANG

Diploma III in Telecommunications Engineering
Tembalang, Semarang, Indonesia
GPA : 3.62 / 4.00

September 2023 – September 2026

SMK NEGERI 1 SLAWI

Computer and Network Engineering
Slawi, Tegal, Indonesia

July 2020 – April 2023

WORK EXPERIENCE

PT TELKOM AKSES

Intern Staff | September 2025 – December 2025
Tegal, Indonesia

- Executed emergency restoration and core-to-core fusion splicing for a 144-core fiber optic feeder cable severed during utility excavation along Jl. Gajah Mada, Tegal.
- Conducted fault location, link loss measurement, and post-repair attenuation verification across reconstructed spans using Optical Time-Domain Reflectometer (OTDR).
- Maintained OLT, ODC, and ODP infrastructure to ensure continuous network stability while documenting technical restoration logs per Telkom Akses standards.

PT TELKOM AKSES

Intern Staff | June 2022 – September 2022
Slawi, Indonesia

- Provisioned and configured Optical Network Terminals (ONT) and Set-Top Boxes (STB) for new IndiHome and IndiBiz subscribers.
- Performed drop-cable deployments, Optical Power Meter (OPM) signal measurements, and validated parameters for IPTV, VoIP, and broadband services.
- Resolved customer-side network incidents (onsite troubleshooting), replacing damaged devices and ensuring SLA installation standards were met.

PROJECT EXPERIENCE

IoT-Based Smart Camera Cabinet with Automated Hysteresis Climate Control

Internet of Things (IoT) & Embedded Systems | 2026

- Built an ESP32-powered camera storage cabinet using a Peltier module, featuring automated humidity regulation, UVC sterilization, and real-time web monitoring.
- Achieved up to 99.21% humidity control accuracy with stable continuous operation exceeding 144 hours.

Microstrip Antenna Design Projects

RF Engineering / Antenna Design | 2025

- Designed and simulated single, 2x2, and 4x1 microstrip antenna configurations for the 902–928 MHz ISM band using CST Studio Suite.
- Evaluated S11, VSWR, gain, directivity, and radiation patterns for sub-GHz applications, achieving optimal S11 around -28 dB and VSWR of 1.1–1.3.

OFDM Wireless Communication Simulator

Wireless Communications / Digital Signal Processing | 2025

- Developed an end-to-end OFDM communication simulator evaluating QPSK, 16-QAM, and 64-QAM modulations over AWGN and Rayleigh channel models.
- Executed FFT/IFFT processing, Zero-Forcing equalization, and BER analysis to evaluate wireless transmission performance across varied SNR conditions.

Radar Target Detection Simulator via Matched Filtering

Digital Signal Processing / Radar Systems | 2025

- Developed a radar signal processing simulator integrating Linear Frequency Modulation (LFM) chirp generation, multi-target echo modeling, AWGN interference, and matched filtering.
- Implemented peak detection algorithms for target range estimation, successfully identifying three simulated targets with zero range estimation error.

ECG Heart Rate Detection & Arrhythmia Analysis System

Biomedical Electronics / Signal Processing | 2026

- Developed an ECG signal processing tool in GNU Octave to filter noise and process raw physiological heartbeat data from the MIT-BIH database.
- Implemented noise filtering and adaptive peak detection to estimate heart rate metrics and classify basic arrhythmia conditions.

Water Resource Monitoring & Billing Management System

Web Application / Smart Water Management System | 2025

- Built a web-based water management application using PHP and MySQL to automate meter recording and customer billing.
- Integrated water usage analytics, customer data management, tariff calculation schemes, and payment history tracking.

CERTIFICATIONS

Radio Frequency Engineer - 193107501110-44/VSGA/BLSDM.Komdigi/2025

Vocational School Graduate Academy (VSGA), Digital Talent Scholarship 2025

Balai Pelatihan Sumber Daya Manusia Komunikasi dan Digital (BLSDM Komdigi) Yogyakarta

CCNA: Switching, Routing and Wireless Essentials

Cisco Networking Academy

CCNA: Introduction to Networks

Cisco Networking Academy

KKNI Level II – Computer and Network Engineering Competency - 61100 3513 2 0000258 2023

Badan Nasional Sertifikasi Profesi (BNSP)

ORGANIZATIONAL EXPERIENCE

KELUARGA MAHASISWA BIDIKMISI POLITEKNIK NEGERI SEMARANG

President | September 2024 – September 2025

- Directed organizational strategy and managed end-to-end execution of academic and leadership programs, building strategic partnerships with internal and external stakeholders.
- Led executive committee members, overseeing budget allocation, project schedules, and performance evaluations for student development initiatives.

HIMPUNAN MAHASISWA ISLAM KOMISARIAT POLINES

General Secretary | June 2025 – Present

- Managed central administrative operations, optimizing official documentation, archive systems, and meeting minutes to ensure compliance.
- Coordinated cross-functional communication among executive members and structured periodic operational reports to enhance organizational transparency.

AWARDS & ACHIEVEMENTS

- 1st Place – Sultan Agung Literacy Fest Essay Competition 2025, Badan Eksekutif Mahasiswa Universitas Sultan Agung
- 2nd Place – Information Network Cabling, Tegal Regency Student Competency Competition 2022, Musyawarah Kerja Kepala Sekolah SMK Kabupaten Tegal
- Silver Medal & Bronze Medal – Pekan Essay Nasional (PENA) 2, Universitas Dhyana Pura
- Best Paper – National Essay Competition 2025, Universitas Negeri Semarang Scholarship Community
- 2nd Runner-Up – Fair Essay Competition 2025, Forum Komunikasi Mahasiswa Politeknik Indonesia, Politeknik Negeri Semarang

TECHNICAL SKILLS

Telecommunications & Networks	: RF & Cellular Networks, Fiber Optic & FTTx, Computer Networking, Routing & Switching, Wireless Networking, Network Troubleshooting
Embedded & Hardware	: ESP32, Arduino, C/C++, IoT Systems, Sensors & Actuators, HTTP/API Communication, MQTT, System Integration
RF & Signal Processing	: RF Engineering, Digital Signal Processing, OFDM, Radar Signal Processing, Matched Filtering, Microstrip Antenna Design
Software & Tools	: Cisco, MikroTik, Cacti, PRTG, Wireshark, Nmap, Atoll, Pathloss, TEMS Investigation, OptiSystem, GNU Octave, Proteus, CST Studio Suite, Microsoft Office
Web & Database	: PHP, JavaScript, HTML/CSS, MySQL

PROFESSIONAL SKILLS

Technical & Operational	: Problem Solving & Technical Troubleshooting, Technical Documentation, Field Operation & Maintenance, System Integration
Management & Soft Skills	: Leadership & Team Coordination, Project Management, Analytical Thinking, Communication & Stakeholder Coordination

LANGUAGES

Indonesian	: Native
English	: Intermediate (POLINES TEPPS Score: 660 Listening: 400, Reading: 260)